

### Question 2:

$$N = \omega \times \frac{\text{pas}}{2\pi} \Rightarrow \omega = N \times \frac{2\pi}{\text{pas}}$$

$$\text{AN: } \omega = 15 \times \frac{2\pi}{2}$$

$$= 15\pi$$

$$\omega \approx 45 \text{ rad/s}$$

### Question 3:

$$C_x = F \times \frac{\text{pas}}{2\pi\eta} \quad \text{Démarrage: } \eta P_e = P_s$$

$$\eta C_x \omega_x = V \times F$$

$$\eta C_x \omega_x = \frac{V}{\pi} \times \frac{\text{pas}}{2\pi} F$$

$$C_x = F \times \frac{\text{pas}}{2\pi\eta}$$

$$\text{AN: } C_x = \frac{300 \times 2 \times 10^{-3}}{2\pi \times 0,5} = \frac{6 \times 10^{-2} \times 2}{2\pi} \approx 0,2 \text{ N.m}$$

$$C_{\text{mot}}(\eta_{\text{max}}) = 0,05 \text{ N.m}$$

$$R = \frac{0,04}{0,2} = \frac{4}{20} = \frac{1}{5} = 0,20$$

$$\text{Vérif de } \omega: C_{\text{pire}}: C_x = \frac{850 \times 2 \times 10^{-3}}{2\pi \times 0,5} = \frac{8,5 \times 2 \times 10^{-1}}{\pi} = \frac{1,7}{\pi} = 0,5 \text{ N.m}$$

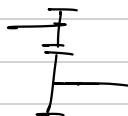
$$C_{\text{mot}}(\text{pire}) = \mu \times 0,5 = 0,1 \text{ N.m}$$

$$\Rightarrow \Omega_{24V}(C_{\text{pire}}) = 2500 \text{ rpm} \Rightarrow \Omega_{\pi}(\text{pire}) = 500 \text{ rpm} \quad V_e(\text{pire}) = 500 \times \text{pas} = \frac{500 \times 2 \times 10^{-3} \text{ m}}{60 \text{ s}} \times \frac{60 \text{ s}}{1 \text{ min}} = \frac{100 \text{ mm}}{6} = 17 \text{ mm/s} > 15 \text{ mm/s}$$

Donc c'est OK

### Question 4

Train simple



9 dents pignon pour minimiser l'encombrement:  $Z_{\text{roue}} = 9 \times 4,8 = 43 \text{ dents}$

$$E = \frac{m(Z_p + Z_r)}{2} = \frac{52 \times 1,25}{2} = 26 + 6 = 32 \text{ mm}$$

$$r_a = \frac{43 \times 1,25}{2} = 27 \text{ mm}$$

$$r_{\text{mot}} = 21 \text{ mm}$$

### Question 5:

$$\text{Encombrement: } 32 + 27 + 21 = 80 \text{ mm} < 82 \text{ mm}$$

Question 7:

$$\phi_{\max} = 50 \text{ mm}$$

$$\phi_{\min} = 20 \text{ mm}$$

$$h = 2,5 \text{ mm}$$

$$\Rightarrow \text{primitif} \triangleq 1 \text{ mm}$$

$$C_t = \int F_P \left( \frac{2 R_{\max}^3 - R_{\min}^3}{R_{\max}^2 - R_{\min}^2} \right) \times 2 \quad \text{surface frottante}$$
$$r_{\text{moy}} = 17,5 \text{ mm} = \frac{35}{2}$$

Question 8:

$$F_P = \frac{C_t}{r_{\text{moy}} \times 2}$$

Question 9: rondelle 0-11850

Question 10: